

Technical talk in bitcoin and equity

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Abstract:

Compared to equity markets, the cryptocurrency market is characterised by high volatility that is commonly without explanation stemming from new information. In this paper, we hypothesise that absent information, investors increase their use of technical trading language to rationalise cryptocurrency price movements. We develop a vocabulary of words designed to capture technical trading language (technical talk) used on large cryptocurrency subreddits on Reddit.com. We find a non-linear relationship between technical talk and bitcoin returns, with technical talk being prominent when bitcoin is performing poorly, and to a lesser extent when it is performing well. The asymmetry in technical talk suggests investors' propensity to search for explanations when assets are performing poorly. We test this hypothesis using quantile regressions, and find technical talk is significantly higher in lower quantiles of bitcoin return. Our results point toward the use of technical talk when information is scarce.

Keywords: *Textual analysis, cryptocurrency, equity markets, social media, Reddit.com*

JEL: G10; G40

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1. Introduction

When asset prices move, there is demand for information to explain those movements. Large volatility in equity markets is seldom without an explanation from the release of new information; and if new information is absent, then volatility is commonly labelled anomalous. However, explanations for the historically high volatility in the cryptocurrency market is less clear. Considerations for the variation in cryptocurrency prices has covered fundamentals and asset pricing factors (Biais, Bisière, Bouvard, Casamatta, & Menkveld, 2023; Liu, Tsyvinski, & Wu, 2022), behavioural (Liu et al., 2022), network effects (Cong, Li, & Wang, 2021; Liu & Tsyvinski, 2021), and links to traditional assets (Yermack, 2015; Baur, Hong, & Lee, 2018). Although high volatility may be explained by the demise of an intermediary or new government regulation (see for example, Conlon, Corbet, & Hu, 2023), such events are infrequent and give irrational or manipulation explanations for frequently high volatility more attention (Gandal, Hamrick, & Oberman, 2018; Liu et al., 2021). If obvious explanations to explain high volatility in cryptocurrency price are scarce, then what alternative explanations might cryptocurrency investors use? In this paper, we investigate whether Bitcoin's performance is associated with technical trading language in popular cryptocurrency subreddits from Reddit.com. We offer a comparison to equity markets, to explore whether technical trading language is more common in environments where information is relatively limited.

There are several reasons to expect the information environment in the cryptocurrency market to be comparatively limited. Equity valuation is commonly linked to macroeconomic and firm specific drivers, which are reported publicly in periodic filings and other disclosures and covered by traditional and social media outlets. The availability of information allows analysts to forecast performance, providing institutional and retail traders with greater information to form opinions on the current value of the asset. In contrast, cryptocurrencies

lack similar connections to macroeconomic trends, have lower regulation requirements, and do not have similar established reporting standards. We propose that as an alternative to comprehensive information, cryptocurrency investors use more technical trading language to explain variations in bitcoin's price.

We refer to technical talk as specialised terminology in financial markets trading. For example, investors referring to declining markets may describe a “bear run”. Furthermore, the terminology can be abbreviated, for example, “*daily simple moving average*” to “*daily sma*”. The challenge in this paper is to create a vocabulary of technical trading language related to financial markets, that is used on Reddit.com, captures abbreviations, and common misspellings. Reddit.com is organised into communities (subreddits), providing a convenient way to capture technical talk used by different communities – in our case cryptocurrency and equity related communities. We use *Word2Vec* to predict similar words to a list of predefined words associated with technical trading in financial markets. On aggregated daily bitcoin returns, we find a U-shaped pattern in technical talk, with technical talk being greatest for very negative returns and to a lesser extent for very positive days. In contrast, we find an inverted U-shape when we search for technical talk in equity related subreddits. The contrast between the cryptocurrency and equity markets supports the notion that large price variation in cryptocurrency markets may be without clear information and therefore increases the use of technical talk; while large variations in equity markets are often explained by the release of information, which reduces the propensity for social media users to use technical talk. We explore the non-linear relationship in quantile regressions and find technical talk is concentrated in lower quantiles of bitcoin's return distribution. We repeat this analysis for an equity market sample and find the relationship between technical talk and S&P 500 returns is negative (positive) in lower (higher) quantiles. In vector auto regressions, we find evidence that a shock to bitcoin returns significantly predicts technical talk. We also find support for the

hypothesis that technical talk is more prevalent when less information is available with shocks to bitcoin returns significantly predicting lower technical talk.

It is possible the asymmetric increase in technical language when bitcoin is performing poorly is associated with loss aversion, whereby investors dislike losses disproportionately more than they like equivalent gains (Kahneman & Tversky, 1979). Furthermore, Festinger's (1957) cognitive dissonance theory, highlights the discomfort experienced by an individual when their choices and beliefs are inconsistent with each other. Chang, Solomon, and Westerfield (2016) argue that when faced with losses, investors are discomforted by their belief of being a good investor colliding with their poor performance. A cryptocurrency investor experiencing poor performance may search for information that confirms their belief that their investment decision is sound and proceed to use technical talk to justify that opinion. Since we assume information in cryptocurrency markets is limited, we expect the propensity for individuals to abandon their belief in a sound investment is also limited. In this context, overconfident investors are prone to attribute poor performance with bad luck (see for example, Barber & Odean, 2002) and technical talk may be seen as a way for investors to confirm their beliefs and restore their perception of being a good trader during bad times.

Our paper contributes to the literature on textual analysis by providing insight into how investors explain asset prices using technical trading language. Understanding this relationship is important given the participation of retail traders, the scarcity of clear explanations for volatility, and how technical language may convince retail investors that poor performance can be explained by using technical language. Textual analysis is used by researchers to capture sentiment, readability, topic modelling, and document similarity (Loughran and McDonald, 2016) from various sources of text such as traditional media, social media, and company filings. A common methodology is to search text using a list of keywords or phrases. For example,

earlier studies search for positive and negative words to measure sentiment (for example, Tetlock 2007; and Garcia, 2013), or use targeted words related to economic policy and uncertainty to create an economic policy uncertainty index (Baker, Bloom, and Davis, 2016). However, searches for keywords in text can be naïve if it fails to account for context, nuances in different languages, or specialized language used in different subjects. In our setting, without a tailored dictionary of words, common abbreviations or misspellings used in subreddit communities could be missed. For example, “hodl” was an unintentional misspelling of “hold”, which subsequently became part of the cryptocurrency culture and represents resilience to “FUD” – another popular crypto-slang which stands for Fear, Uncertainty, and Doubt. To capture such linguistic nuances would be difficult through manual investigation of millions of comments.

Overcoming this challenge is made easier by the development and accessibility to machine learning algorithms. For example, Du, Huang, Wermers, and Wu (2022) highlight the difficulty in capturing subtleties and meanings of certain words between languages and use *Word2Vec* to construct a sentiment dictionary on Chinese financial news. The importance of doing this is highlighted by the authors finding positive news tone has a greater impact on returns compared with negative tone, a result that is contrary to Tetlock, Saar-Tsechansky, and Macskassy (2008) and Huang, Tan, and Wermers (2020), among others. Nasekin and Chen (2020) use different machine learning methods to capture cryptocurrency sentiment in StockTwits messages, acknowledging the importance of capturing specialized language used in a social media setting (e.g., the use of emojis and emoticons). Similarly, Sautner, Ven Lent, Volkov, and Zhang (2023) use an algorithm to capture keywords related to firms’ exposure to climate change, highlighting that libraries or predefined keyword lists may do a poor job at capturing specialized language used in the context of climate change.

2. Data

We use Reddit data from Pushshift’s API (see Baumgartner, Zannettou, Keeban, Squire & Blackburn, 2020).¹ We expect that the reference to technical trading descriptions to differ between traditional media and social media. Since we are analysing social media comments, it is appropriate to consider a vocabulary that captures technical talk in this context. Using *Word2Vec*, we train a neural network on comments from a popular trading community on Reddit (r/daytrading)². Word2Vec is a neural network designed to capture semantic meaning by assuming words that appear in common contexts will have similar meanings. The algorithm begins by building a vocabulary from inputted text to generate word vectors for each word in the vocabulary. Using a list of pre-defined target words (Table 1), the trained model predicts words in r/daytrading comments with a similar context and determines the cosine similarity between the target word and each word in the vocabulary. We filter the predicted words by taking the 100 most similar words predicted by the model for each target word. Combining our list of target words with words predicted by the model gives us 3,075 unique words. We intervene by removing words that are ambiguously or not explicitly related to technical trading language. For example, “previous period” can relate to technical trading in the given context, it is also reasonable to expect it to be used in other contexts related to (for example) Bitcoin’s blockchain. We remove 1,719 words, which gives us a final sample of 1,356 technical talk keywords. In Appendix Table A2, we show a sample of the predicted words in our keyword list (Panel A) and predicted words that were removed (Panel B).

[Insert Table 1 about here]

¹ Pushshift’s Reddit API was made unavailable in early 2023.

² As of May 2023, r/daytrading had 1.5 million members.

We use approximately 44 million comments between 2017 and 2022 from two of the largest cryptocurrency subreddits on Reddit.com – r/Bitcoin and r/CryptoCurrency.³ For the remainder of the paper we refer to these as cryptocurrency subreddits for brevity. We search comments for technical talk keywords and assign a value of one if at least one keyword appears in the comment, and zero otherwise. We aggregate the comment data by counting the number comments flagged as containing technical talk at a daily frequency. Because variations in daily technical talk is likely to be a function of total activity on the subreddits, we calculate *Crypto Ratio* as the number of technical talk comments on day t divided by the total number of comments on day t . We repeat this procedure, to construct *Equity Ratio* as the number of technical talk comments from equity-related subreddits and explore the relationship with S&P 500 returns in later analysis. We collect and merge daily bitcoin prices on Bitstamp from Refinitiv Eikon with our aggregated comment data.

[Insert Figure 1 about here]

In Figure 1, we plot the 30-day moving average in the *Crypto Ratio* and bitcoin returns. Bitcoin returns are calculated as the natural log of bitcoin price on day t divided by the bitcoin price on day $t-1$. Similar to bitcoin's historical volatility, there is substantial variation in the use of technical talk in the cryptocurrency subreddits. The largest proportion of technical talk was used during 2022, with approximately 4.5 per cent of the total comments containing technical talk. We observe an inverse relationship between bitcoin's 30-day moving average and our technical talk ratio, suggesting that technical talk increases (decreases) relative to total comments when bitcoin returns are decreasing (increasing).

³ As of September 2023, there were 5.8 million members on r/Bitcoin and 6.8 million members on r/CryptoCurrency.

Table 2 shows the descriptive statistics for our daily sample between 2017 and 2022. The proportion of technical talk on cryptocurrency subreddits (*Crypto Ratio*) is similar to equity subreddits (*Equity Ratio*). The mean of *Crypto Ratio* (3.1%) is closely aligned with that of *Equity Ratio* (3.2%), both showing a similar percentage of daily comments that contain at least one reference to technical trading language. The median and maximum values of *Crypto Ratio* closely resemble those of *Equity Ratio*. This suggests a comparable frequency of technical jargon usage across these subreddits.

[Insert Table 2 about here]

Table 3 presents correlations of key variables. *Crypto Ratio* and *Equity Ratio* exhibits a negative correlation of -0.231. This suggests an inverse contemporaneous relationship between the intensity of technical discussion on cryptocurrency and equity-focussed subreddits. The weak negative correlation (-0.082) between daily bitcoin returns and daily comments on cryptocurrency subreddits implies a tendency for more (fewer) comments to be made when Bitcoin prices are lower (higher). This provides preliminary evidence that technical talk relative to total comments increases when bitcoin prices are lower. The weak negative correlation between *Crypto Ratio* and equity market return (-0.0049) indicates minimal inverse association between technical talk on cryptocurrency subreddits and mainstream equity market movements. In contrast, there is a weak positive correlation (0.0032) between *Equity Ratio* and S&P 500 returns (*SP500*).

[Insert Table 3 about here]

3. Results

3.1 Positive versus negative return days

In this section we explore the use of technical talk depending on the performance of bitcoin. Since Bitcoin is considered a speculative asset and explanations for variations in its price are often absent fundamental information, we expect that users on social media will rely more on technical talk on the tail ends of bitcoin's return distribution. We sort our sample into deciles based on bitcoin's daily return, where decile one (ten) contains the smallest (largest) daily returns. In Figure 2, we plot the average number of comments *and Crypto Ratio* by bitcoin return decile.

[Insert Figure 2 about here]

We find a U-shaped pattern in both the number of comments and technical talk. On average, the number of comments is largest when bitcoin is performing poorly or well. Since large positive and negative returns are likely to be more attention grabbing, it is unsurprising that activity on the cryptocurrency subreddits increases. There is a similar pattern in *Crypto Ratio*. High and low return deciles correspond with a greater proportion of technical talk. While large positive and negative returns garner higher attention, the asymmetry observed in technical talk suggests a tendency for users to explain poor performance with more technical language.

3.2 Technical talk and bitcoin return regressions.

In this section, we investigate the non-linear relationship observed in Figure 2 using quantile regressions, while controlling for Bitcoin network variables, market factors, economic political uncertainty, and gold prices. Network variables are found to predict cryptocurrency returns (Liu and Tsyvinsky, 2021), we perform a principal component analysis (PCA) and use

the first component as composite Bitcoin network index (*PCI Network*). We include the following network variables from blockchain.com in the PCA: number of transactions, number of unique addresses, estimated transaction value, unspent bitcoin transactions, number of transactions per second, mempool size, and mempool count. We provide definitions for these variables in the appendix (Table 1A).

Tables 4 and 5 show the eigenvalues and variance explained by each component, and the correlation between components and variables, respectively. The correlation between the first component and the percentage change in transactions, unique addresses, transaction volume, and transactions per second are 46.5%, 48.9%, 41.6%, and 47.0%, respectively. Therefore, the first component appears to capture variation in transaction related variables, while the second component has high correlations with the percentage change in mempool size and mempool count, 62.2% and 61.8%, respectively. There is a 94.3% correlation between percentage change in unspent transactions and the third component.

[Insert Table 4 about here]

[Insert Table 5 about here]

The remaining control variables are daily percentage change in economic political uncertainty (*ΔEPU*), and the daily change in the gold spot price (*ΔGold*). Additionally, we include size, value, investment, and profitability factors (Fama and French, 2015) from Kenneth French's data library⁴. Since we include SP&500 returns (*SP500*) in our regressions, we omit the market factor to avoid collinearity.

⁴ https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html

In Table 6, we find a significantly negative contemporaneous relationship between daily bitcoin returns the proportion of technical talk on cryptocurrency subreddits. We find this relationship persists after including several controls. To test for the non-linear relationship, we use simultaneous quantile regressions to evaluate how technical talk is associated with bitcoin returns. We estimate this association over five quantiles 10%, 30%, 50%, 70%, and 90%. Results are shown in Table 7. At the 10th percentile (Q10), higher *Crypto Ratio* is associated with lower bitcoin returns with a coefficient of -1.315, and the result is statistically significant at the 0.1% level after including all control variables. This suggests that when bitcoin returns are at their lower extremes and the market is performing poorly, an increase in technical discussion correlates strongly with these lower returns. A similar result is found at the 30th percentile, which also sees a significantly (at the 1% level) negative impact from *Crypto Ratio*, albeit the magnitude of the effect is less than at the Q10. This implies that among low, whilst not extreme daily bitcoin returns, an increase in technical discussion is still associated with lower returns but the association is weaker compared to the Q10. At the median, the relationship becomes weaker with a coefficient of -0.217, which is not statistically significant at the 5% level. This shows that around the median daily bitcoin returns, the impact of *Crypto Ratio* is minimal and not statistically distinguishable from zero. When we move to upper quantiles, the coefficient nears zero for Q70 (-0.014) and slightly positive for Q90 (0.121), however, these relationships are not statistically significant at the 5% level.

Overall, these results are supportive of a general trend of a non-linear relationship that varies across the distribution of returns, as the relationship between *Crypto Ratio* and daily bitcoin returns becomes less negative as we move from the lower to the upper deciles of bitcoin's return distribution. These findings suggest that the content of discussions may be a more crucial factor during bitcoin market downturns. The trend is in line with the U-shaped curve illustrated in Figure 2.

[Insert Table 6 about here]

[Insert Table 7 about here]

3.3 Equity market comparison

In contrast to cryptocurrency markets, variations in equity valuation are commonly explainable by new fundamental information. In fact, market volatility without explanation is regarded anomalous or irrational. In this section, we use the same list of technical trading words to search approximately 74 million comments from large equity-market related subreddits – r/wallstreetbets and r/stocks. We compare the use of technical talk on these subreddits with returns on the S&P 500 index. Similar to the previous section, we sort our sample into deciles based on daily S&P 500 returns, where decile one (ten) is the average of the lowest (highest) daily returns in our sample. Unlike the observation in the bitcoin sample, Figure 3 is n-shaped in average technical talk by S&P 500 return decile. Specifically, technical talk is lowest at high and low S&P 500 return deciles, and highest in the middle deciles. We expect that the demand for technical information on days when the S&P 500 experiences higher volatility is lower since large variations in equity prices are commonly associated with the release of new information. A larger proportion of technical talk around the middle return deciles is consistent with the notion that users on social media attempt to predict future movements. This is in contrast with the Figure 2, which suggests technical talk may be adopted to explain significant bitcoin price movements ex-post.

[Insert Figure 3 about here]

We perform similar OLS regressions as in the previous section and test the non-linear relationship between technical talk in equity-related subreddits and S&P500 returns. In Table

8, the OLS estimates present no significant relationship between SP500 returns and *Equity Ratio*.

To test the non-linear relationship, we perform a similar analysis using simultaneous quantile regressions. Results are shown in Table 9. At the 10th percentile (Q10), the positive and statistically significant (at 0.1%) coefficient (0.283) suggests that when S&P 500 returns are in the lower tail, an increase in technical discussion is associated with an increase in returns. This relationship continues to be positive and significant when we move to the 30th percentile (Q30), though the magnitude of this effect decreases. At the median level (Q50), the impact from *Equity Ratio* is almost negligible (0.002) and not statistically significant at the 5% level. When we move to upper quantiles Q70 and Q90, we see a statistically significant (at 0.1%) negative relationship with the *Equity Ratio*, with coefficients of -0.0095 and -0.170, respectively. This indicates that in periods of high market returns, an increase in technical jargon is associated with a decrease in returns. The trend is consistent with the n-shaped curve illustrated in Figure 3.

Similar to the earlier analysis on the bitcoin market, these results are also supportive of a non-linear relationship, albeit with different direction. For Bitcoin, the relationship with *Crypto Ratio* was mostly negative across quantiles and strongest in lower quantiles of bitcoin's return distribution. In contrast, *Equity Ratio* shows a positive relationship in lower quantiles and turns negative in upper S&P 500 return quantiles. The significance at extreme quantiles is notable, implying that technical discussion has a greater impact during extreme equity market conditions.

These differences may reflect the varying sensitivities of the two markets to discussion involving technical talk and support the view that different investor behaviours are observed in cryptocurrency and equity markets. While this analysis only highlights a contemporaneous relationship, it is possible cryptocurrency investors react to social media content with more

technical talk in times of uncertainty or negative trends to confirm prior beliefs that their cryptocurrency investment is sound. Equity investors might find technical discussion reassuring in downside and neutral market conditions but become cautious of technical discussions when the market is performing well. Overall, these results emphasise the importance of market context in understanding the relationship between technical discussion and asset returns.

[Insert Table 8 about here]

[Insert Table 9 about here]

3.4 Cryptocurrency news and technical talk

So far, we have assumed that technical talk is prevalent in cryptocurrency subreddits because significant volatility in bitcoin returns is often absent obvious information. If this hypothesis holds, we expect that technical talk will be relatively lower when reasons for price volatility are obvious. We proxy the amount of information circulating on the cryptocurrency subreddits by using references made to external media outlets.

We flag submissions made on r/Bitcoin and r/CryptoCurrency with one if it contains a URL to a media outlet. Since cryptocurrency and traditional media outlets may have differences on how they report on Bitcoin, we separate references to *Traditional* media and *Cryptocurrency* media. Table 10 provides a list of domain names we use to identify sources of information.

[Insert Table 10 about here]

Since we do not investigate the content of referenced media, we assume the provision of an external media source refers to newsworthy or attention-grabbing media articles related to cryptocurrencies. To strengthen this assumption, we use the number of comments made on submissions referencing news articles to give more (less) weight to referenced news that receives more (less) attention. Similar to the construction of our technical talk ratio, we standardise the number of comments from submissions with references to URL domains listed in Table 10 on day t , by the total number of comments made across all submissions on day t .

An issue with using a proportion is that we may be capturing an attention effect. Given that media outlets are not expected to significantly increase their publication of articles on days with abnormal Bitcoin volatility, the volume of news could be dominated by heightened investor attention. Specifically, on such days, the frequency of news articles shared by Redditors might not rise in tandem with the surge in attention. This is because a single article submission post is less likely to be repeatedly referenced within the same subreddit on the same day. Consequently, to capture this dynamic more accurately, we consider the abnormal number of comments on submissions referencing a media source as an alternative proxy for gauging information flow. We define abnormal news as:

$$Abnormal\ News_t = \ln(1 + News\ Urls_t) - \ln(1 + \overline{News\ Urls_t})$$

$$\text{where, } \overline{News\ Urls_t} = \frac{\sum_{n=-30}^{n=-1} News\ Url_{t,n}}{30}$$

Where $Abnormal\ News_t$ is the abnormal number of comments from submissions referencing a media source on day t . $News\ Urls_t$ is the number of comments from submissions referencing a media source on day t . $\overline{News\ Urls_t}$ is the 30-day moving average of the number of comments from submissions referencing a media source on day t .

In Table 11, we run quantile regressions, with the proportion of news (*Crypto News* and *Traditional News*) and abnormal news (*AbnCrypto News* and *AbnTraditional News*) as the dependent variables.

From Q10 to Q90 of *Crypto News*, the coefficients of *Crypto Ratio* range from 0.024 to -0.0135. All coefficients are not statistically significant at the 5% level, indicating that there is no strong evidence to suggest that *Crypto Ratio* consistently affects *Crypto News* across the distribution. In comparison, the coefficients on abnormal cryptocurrency news (*Abn Crypto News*) decrease from 4.684 to -13.166 across quantiles. In particular, the negative coefficients at Q50, Q70 and Q90 are all statistically significant at the 1% level. This suggests a strong negative relationship at the median and upper end of the distribution, indicating that higher *Crypto Ratio* is associated with a lower count of abnormal crypto news, whereas there is no clear relationship at the lower end of the distribution. This highlights, that technical talk decreases more when there are high levels of news compared to when news is relatively scarce.

For *Traditional News*, while we see a significantly positive relationship at the lower quantiles and median, the relationship is not clear in the upper end of the distribution. We see a similar relationship in *Abn Traditional News* to that of *Abn Crypto News*, where coefficients of *Crypto Ratio* ranges from 18.561 at Q10 to -15.766 to Q90. Coefficients are statistically significant at the lower end as well as at the higher end of the distribution. After aggregating the number of comments from submissions referencing news from cryptocurrency and traditional media outlets (*Total News* and *Abn Total News*), we find the relationship between total news and technical language is only significantly negative in the highest quantile (Q90). It is worth highlighting that the correlation between *Abn Crypto News* and *Abn Traditional News* is only 8.13%, suggesting the attention paid to each source of information are reasonably independent.

Overall, our results highlight the relationship between technical language and the level of news, is most negative when comments on submissions referencing news is high. This supports our hypothesis that users on the subreddit forums have less reliance on technical language to explain bitcoin price volatility when information is abundant.

[Insert Table 11 about here]

3.5 VAR – Cryptocurrency news, technical talk, and bitcoin returns.

So far, we have explored the contemporaneous relationship between cryptocurrency performance technical talk. In this section, we use vector auto regression (VAR) analysis to test the lead lag relationship between bitcoin returns, technical talk, and cryptocurrency news. If bitcoin investors increase the use of technical language to explain bitcoin's performance, then we expect to observe a significant change in *Technical Talk* to a shock in *Bitcoin* returns. We use the abnormal number of comments from submissions referencing news articles (*Abn Crypto News* and *Abn Traditional News*) in our VAR setup, since we find it to be more effective at explaining the contemporaneous relationship with technical language (*Crypto Ratio*) in the quantile regressions.

We standardise *Technical Talk*, *Abn Crypto News*, *Abn Traditional News*, and *Bitcoin* returns to have a mean of zero and a standard deviation of one. This transformation allows us to interpret the VAR coefficients as the impact of a one standard shock in the dependent variable. In Figure 4, we show orthogonalized impulse response functions for a five-equation VAR setup including *Bitcoin* returns, *Crypto Ratio*, *Abn Crypto News*, *Abn Traditional News*, and *PC Network* as endogenous variables, and *SP500*, ΔEPU , and $\Delta Gold$ as exogenous variables. We use five lags for each endogenous variable, which is determined as the optimal lag length using Schwarz's Bayesian information criterion. Since we are mainly interested in

the relationship between bitcoin returns, news, and the use of technical language, we do not include the impulse response functions for our bitcoin network variable (*PC Network*).

In Panel A, we show the response of *Crypto Ratio*, *Crypto News*, and *Traditional News* to a one standard deviation shock in *Bitcoin* return. A shock in bitcoin returns is associated with significantly lower *Crypto Ratio*. The negative response in technical language highlights that Bitcoin investors increase the amount of technical talk following days of poor bitcoin performance. It is worth noting that our analysis does not differentiate technical language used to explain poor performance from users expressing future expectations for bitcoin's price. Two explanations can be provided. We could hypothesise users on social media use technical language to explain poor performance when information is scarce. Alternatively, our measure of technical language captures users' reasoning for possible reversals if bitcoin has recently decreased. In either scenario, we argue technical language is used more given the scarcity of information. Overall, the negative relationship observed in Panel A supports earlier findings of the negative relationship between contemporaneous Bitcoin performance and the amount of technical language used on social media. We do not find evidence that cryptocurrency or traditional news changes significantly from a shock in bitcoin returns. In Panel B, we find a shock to *Crypto Ratio*, has a mostly insignificant effect on *Bitcoin*, *Crypto News*, and *Traditional News*. There is some evidence of the amount of technical language predicting a significant increase in bitcoin return, however, this effect reverses and becomes insignificant in the following days.

Panels C and D show the cumulative impulse response functions for shocks to *Crypto News* and *Traditional News*, respectively. If technical language is a substitute to describe bitcoin volatility when information from other sources is scarce, then we expect a shock to news will decrease the use of technical language. The significantly negative response in *Crypto Ratio* to a shock in *Crypto News* (Panel C) is seen for the first four days before turning

insignificant. While this supports our hypothesis, the effect is positive but insignificant for a shock to *Traditional News* (Panel D).

[Insert Figure 4 about here]

In Figure 5, we show orthogonalized cumulative impulse response functions for the equity sample, using a three-equation VAR system that includes *SP500*, *Equity News*, and *Equity Ratio* as endogenous variables, and ΔEPU and $\Delta Gold$ as exogenous variables. Three lags are determined as optimal by Schwarz's Bayesian information criterion. We observe a shock to S&P 500 returns does not predict lower levels of technical talk on equity subreddits (Panel A). This is in contrast with a shock to bitcoin returns predicting lower technical talk in cryptocurrency subreddits. It is possible that the contrasting results highlights the difference in the availability of information and the propensity for users to use technical talk when faced with shocks to market returns. We do not observe any predictability in equity-related news or S&P 500 returns from a shock to the amount of technical talk (Panel B). In Panel C, we find a shock to the amount of equity news, significantly predicts a lower *Equity Ratio*. This is a similar reaction observed in the cryptocurrency sample (Figure 4, Panel C), and suggests the need for technical talk is reduced when there is more information.

[Insert Figure 5 about here]

4. Conclusions

In this paper, we define a list of words tailored to capture technical trading language that includes abbreviations and misspellings in the context of social media platform, Reddit.com. We refer to the use of technical language as technical talk. Assuming the availability of information in the cryptocurrency market comparatively worse to traditional

markets, we hypothesise that cryptocurrency investors will have a propensity to use technical language to explain high volatility. We use comments from two popular cryptocurrency subreddit communities (r/Bitcoin and r/CryptoCurrency) on Reddit.com and find technical talk increases during bad times, and to a lesser extent in good times. In quantile regressions, we find this negative relationship between technical talk and bitcoin performance is limited to the lowest quantiles of bitcoin's return distribution in our sample. The results highlight that bitcoin's historical volatility may commonly be unaccompanied by the release of new information. There appears to be a tendency for investors to explain poor performance in environments with comparably less information by using technical language to dissuade interpretations of bad performance.

To study this further, we offer a comparison to the S&P 500 returns and technical talk on two large equity-related subreddits (r/wallstreetbets and r/stocks). Since equity markets are characterised by more information and regulations for investors to explain performance, we the use of technical language to differ compared to cryptocurrency markets. We find that technical talk is lowest (highest) on the tails (middle) of the S&P 500 return distribution. For equity markets to move significantly, it is expected for there to be new information that has a material impact on forecast of future cash flows and/or perception of risk. In similar vector auto regressions, we find no evidence to suggest S&P 500 returns can predict technical talk, while a shock to news decreases technical talk.

Our paper adds to the literature on textual analysis by using machine learning methodology to capture technical trading language in social media posts. The findings provide an interesting insight into the use of specific language on social media and provides room for future research to explore whether technical talk can convince or alleviate the negative emotions investors experience during bad times.

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Figure 1: Technical Talk Ratio and Bitcoin Returns (30 day Moving Average)

This figure shows the 30-day moving average of technical talk ratio and bitcoin returns. Technical talk is calculated as the number of comments containing technical talk on day t divided by the total number of comments on day t .

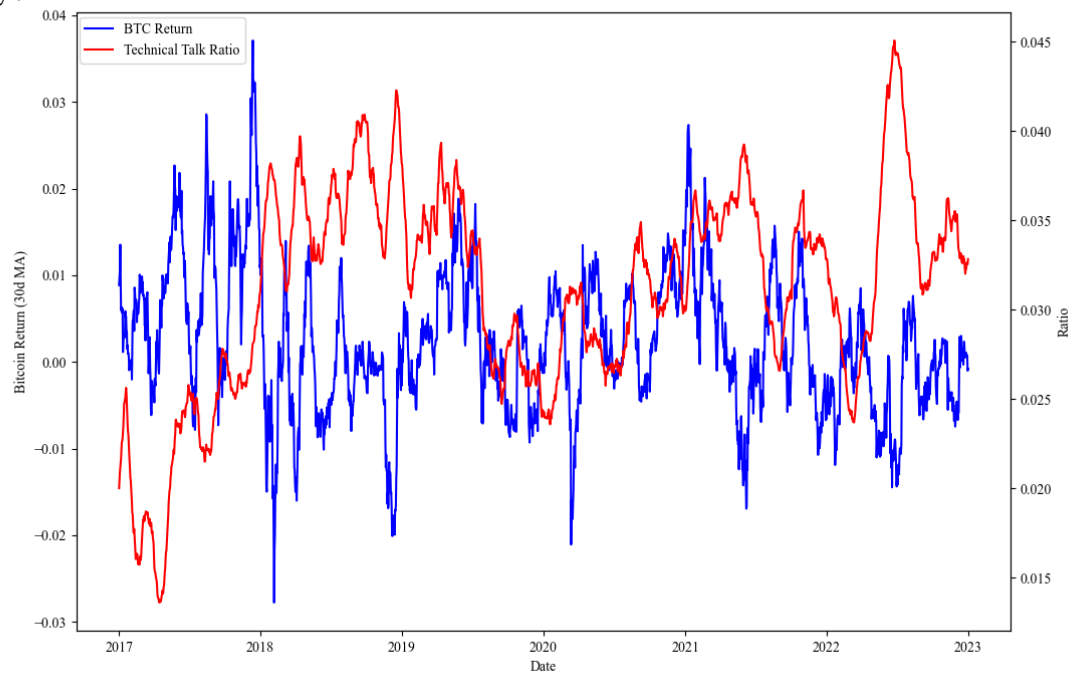


Figure 2: Technical Talk Ratio and Comment Count by bitcoin return decile

This figure shows the average Technical Talk Ratio and average comment count from large cryptocurrency subreddits by bitcoin return deciles. Daily bitcoin returns are sorted into deciles where deciles 1 (10) is the lowest (highest) bitcoin returns in the sample.

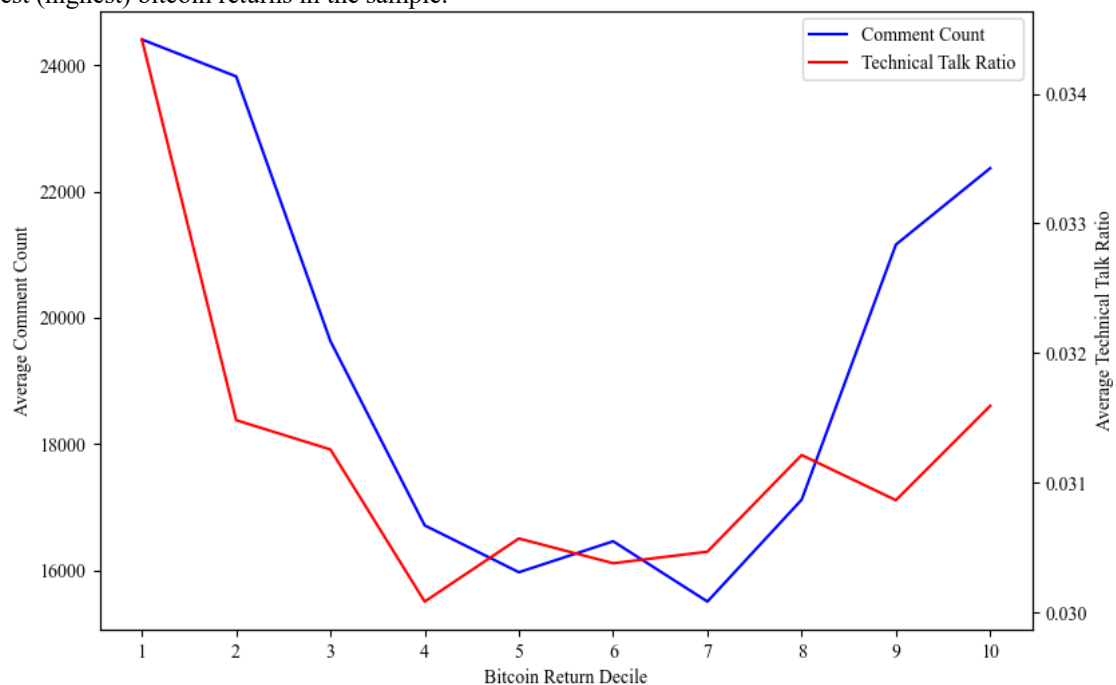


Figure 3: Technical Talk Ratio and Comment Count by S&P 500 return decile

This figure shows the average Technical Talk Ratio and average comment count from large equity market related subreddits by S&P 500 return deciles. Daily S&P 500 returns are sorted into deciles where deciles 1 (10) is the lowest (highest) S&P 500 returns in the sample.

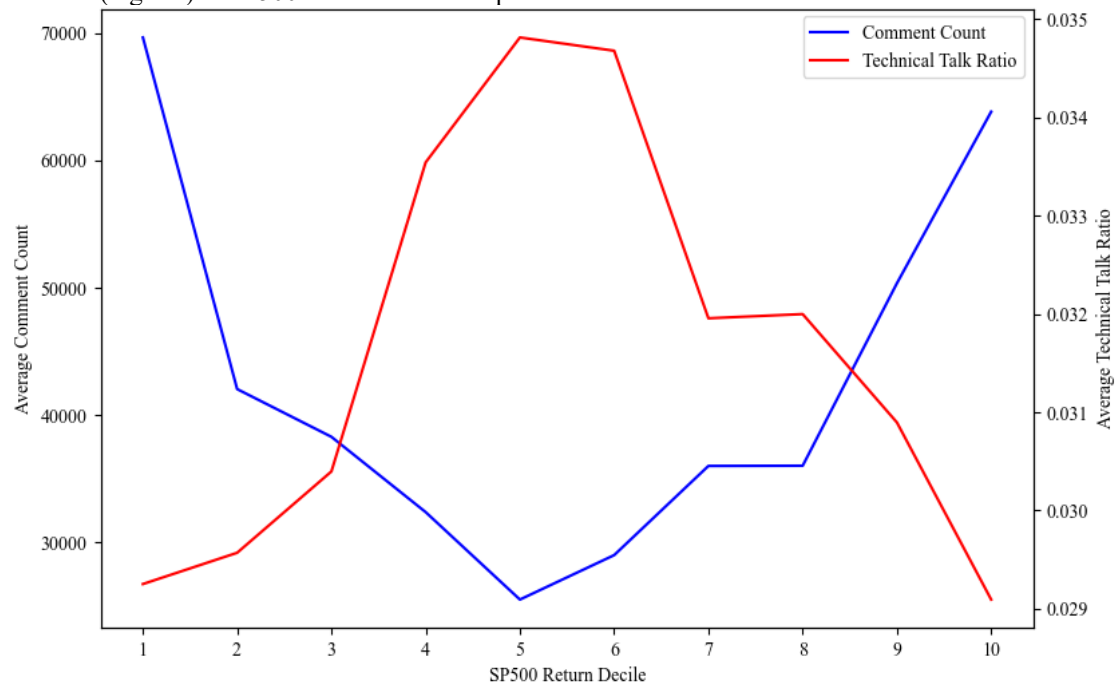


Figure 4: Cumulative Impulse Response Functions – Cryptocurrency sample

This figure shows the orthogonalized cumulative impulse response functions (IRF) for *Bitcoin*, *Crypto Ratio*, and *Crypto News*, using a five equation VAR with five lags for each endogenous variable and *SP500*, ΔEPU , and $\Delta Gold$ as exogenous variables. The number of lags is determined using Schwarz's Bayesian information criterion (BIC). The solid line shows the orthogonalized cumulative impulse response from a one standard deviation shock. The dashed lines are the 95% confidence interval bands and are calculated from bootstrapped standard errors using 1,000 replications.

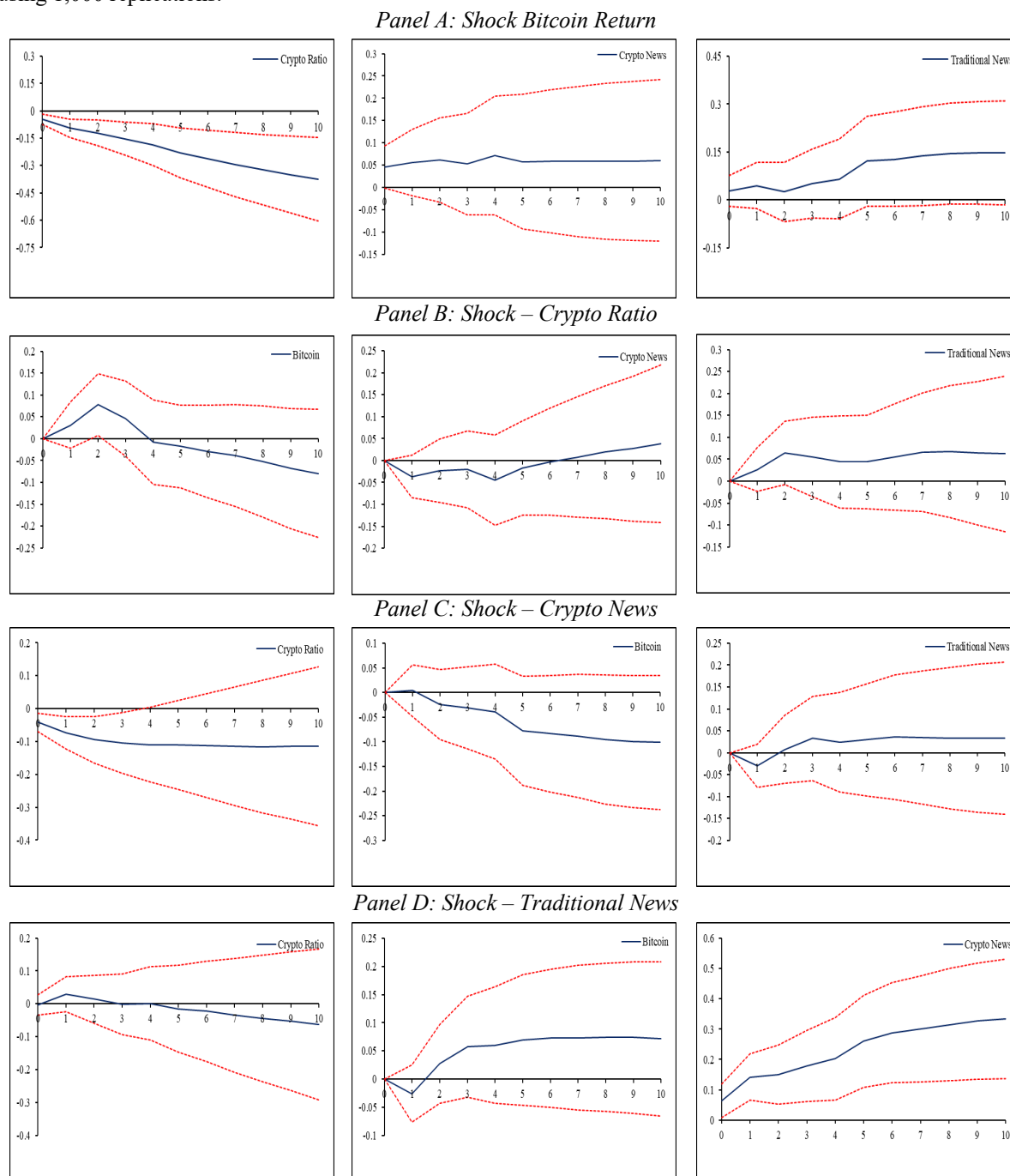


Figure 4: Cumulative Impulse Response Functions – Equity sample

This figure shows the orthogonalized cumulative impulse response functions (IRF) for *SP500*, *Equity Ratio*, and *Equity News*, using a three equation VAR with three lags for each endogenous variable and ΔEPU , and $\Delta Gold$ as exogenous variables. The number of lags is determined using Schwarz's Bayesian information criterion (BIC). The solid line shows the orthogonalized cumulative impulse response from a one standard deviation shock. The dashed lines are the 95% confidence interval bands and are calculated from bootstrapped standard errors using 1,000 replications.

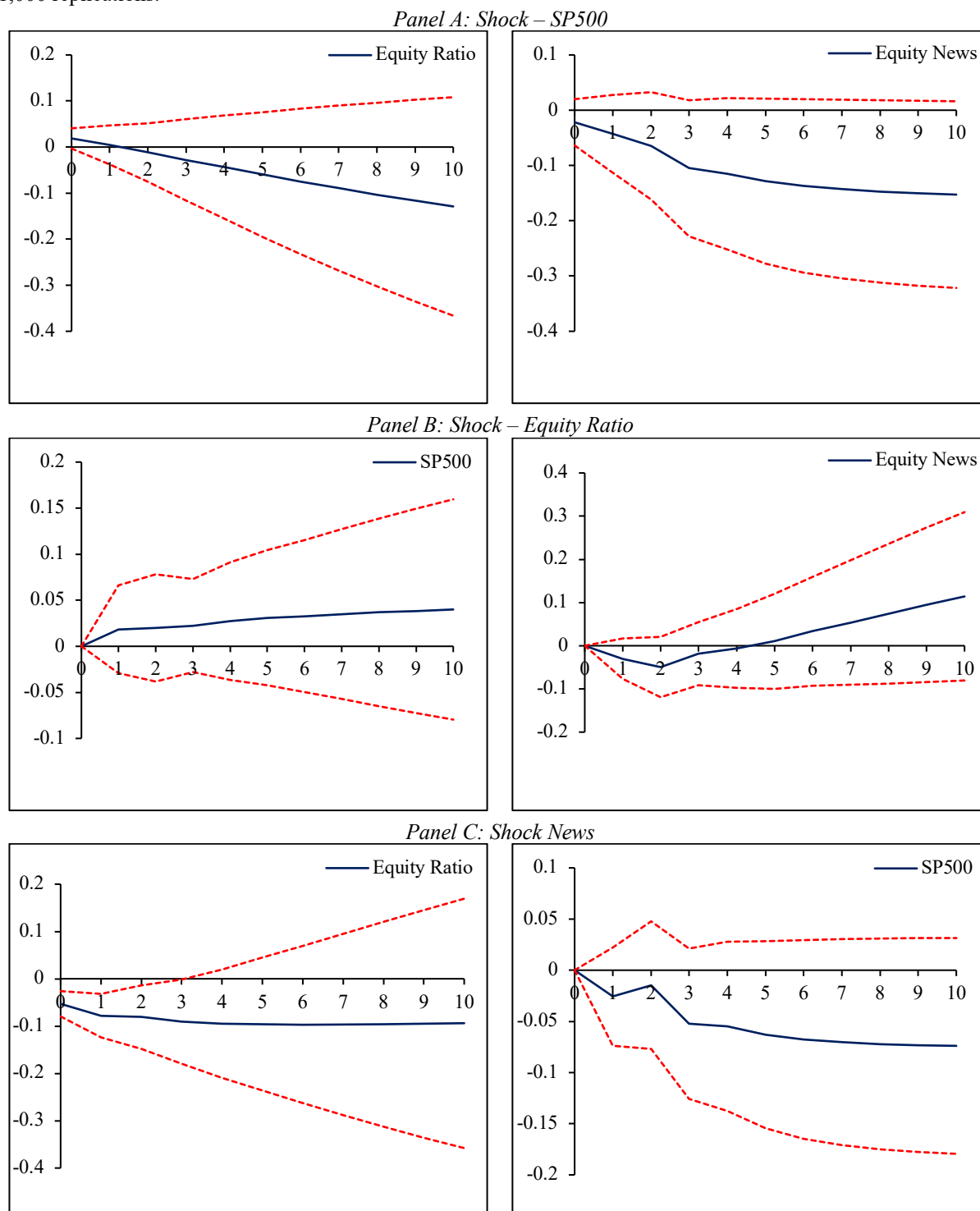


Table 1: Target words

absolute change	correlation coefficient	overvalued	technical analysis
accumulation distribution	cryptocurrency indicator	parabolic sar	time series
accumulation distribution volume	cumulative return	polynomial regression	trading volumes
accumulation phase	directional index	price channel	trend strength
alexander's filter	directional indicator	price index	undervalue
all time high	ease of movement	price momentum	undervalued
all time low	exponential regression	price momentum	volatility index
alpha indicator	head and shoulders	price rebasing	volume index
alpha trend	hedge ratio	price volume	volume momentum
arbitrage	hit the bottom	rainbow chart	volume price
aroon	Ichimoku	regression analysis	wycoff
average price	indexed price	regression band	
average true range	kairi	regression graph	
bar chart	keltner	regression model	
bear run	limit order	regression projection	
bearish	line chart	regression slope	
beta indicator	linear regression	relative performance	
beta trend	logarithmic regression	relative strength compare	
bid ask	macd	relative strength index	
bollinger band	market cap	relative volatility	
bollinger bands	market indicator	rsi divergence	
bull run	market order	rsi indicator	
bullish	median price	rsi moving average	
butterfly spread	momentum index	running sum	
candlestick	money flow index	standard deviation	
chaikin	moving average	standard error	
channel resistance	ohlc	stochastic indicator	
commodity channel index	on balance volume	stochastic momentum	
conversion premium	oscillator	swing index	
correlation	overvalue	swing index	

Table 2: Descriptive statistics

This table shows the descriptive statistics. Variable definitions are in Appendix Table A1.

	Obs	Mean	Median	Std	Min	Max
<i>Crypto ratio</i>	1,510	0.031	0.031	0.008	0.007	0.069
<i>Equity ratio</i>	1,510	0.032	0.029	0.009	0.006	0.070
<i>Bitcoin</i>	1,510	0.002	0.002	0.045	-0.494	0.238
<i>SP500</i>	1,510	0.000	0.001	0.013	-0.128	0.090
<i>ΔEPU</i>	1,510	0.000	-0.060	0.082	-0.300	0.768
<i>ΔGold</i>	1,510	0.000	0.000	0.009	-0.051	0.056
<i>PCI Network</i>	1,510	0.000	-0.395	1.822	-8.560	7.293

Table 3: Correlations

This table shows the descriptive statistics. Variable definitions are in Appendix Table A1.

	<i>Crypto ratio</i>	<i>Equity ratio</i>	<i>Bitcoin</i>	<i>SP500</i>	<i>ΔEPU</i>	<i>ΔGold</i>
<i>Equity ratio</i>	-0.231					
<i>Bitcoin</i>	-0.082	0.016				
<i>SP500</i>	-0.049	0.032	0.270			
<i>ΔEPU</i>	-0.016	-0.009	0.007	0.045		
<i>ΔGold</i>	-0.045	0.002	0.097	0.073	-0.003	
<i>PCI Network</i>	0.075	-0.037	0.034	-0.021	-0.129	-0.001

Table 4: Principal Component Analysis results

This table shows the eigenvalues from the PCA on change in daily bitcoin transactions, unique addresses, transaction volume, unspent transactions, transactions per second, mempool size, and mempool count.

	Eigenvalue	Variance explained	Cumulative
<i>Component 1</i>	3.320	0.474	0.474
<i>Component 2</i>	1.666	0.238	0.712
<i>Component 3</i>	0.933	0.133	0.846
<i>Component 4</i>	0.526	0.075	0.921
<i>Component 5</i>	0.286	0.041	0.962
<i>Component 6</i>	0.166	0.024	0.985
<i>Component 7</i>	0.103	0.015	1.000

Table 5: Component and factor correlations

This table shows the factor loadings from the PCA on daily change in Bitcoin network variables.

	Component 1	Component 2	Component 3	Unexplained
<i>ΔTransactions</i>	0.465	-0.280	0.187	0.120
<i>ΔUnique addresses</i>	0.488	-0.169	0.155	0.141
<i>ΔTransaction volume</i>	0.416	-0.016	-0.007	0.425
<i>ΔUnspent transactions</i>	-0.079	0.294	0.943	0.007
<i>ΔTransactions perSecond</i>	0.470	-0.197	0.003	0.202
<i>ΔMempool size</i>	0.269	0.622	-0.132	0.100
<i>ΔMempool count</i>	0.271	0.618	-0.187	0.087

Table 6: Crypto Ratio OLS regressions

This table shows the association between daily bitcoin returns and technical talk on cryptocurrency subreddits. The dependent variable is the natural log of bitcoin price on day t divided by bitcoin price on day $t-1$. *Crypto Ratio* is the number of comments that contain at least one technical word on day t divided by the total number of comments on day t . Other variables are defined in Appendix Table A1. Coefficients are estimated using Newey-West standard errors. t statistics in parentheses * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

	(1)	(2)	(3)	(4)
<i>Crypto Ratio</i>	-0.485** (-3.03)	-0.503** (-3.20)	-0.405** (-2.71)	-0.410** (-2.75)
<i>PCI Network</i>		0.001 (1.37)	0.001 (1.66)	0.001 (1.68)
<i>ΔEPU</i>			0.000 (0.02)	-0.001 (-0.21)
<i>ΔGold</i>			0.369** (2.75)	0.323* (2.42)
<i>SP500</i>			0.932*** (4.80)	0.860*** (3.94)
<i>SMB</i>				0.465* (2.29)
<i>HML</i>				-0.224 (-1.40)
<i>RMW</i>				-0.354 (-1.52)
<i>CMA</i>				-0.198 (-0.55)
<i>Constant</i>	0.018*** (3.41)	0.018*** (3.56)	0.014** (2.91)	0.015** (3.02)
<i>Quarterly FE</i>	Yes	Yes	Yes	Yes
<i>Adj R²</i>	0.004	0.005	0.080	0.091
<i>Obs</i>	1,510	1,510	1,510	1,510

Table 7: Crypto ratio quantile regressions

This table shows the quantile regression results on the association between daily bitcoin returns and technical talk on cryptocurrency subreddits. The dependent variable is the natural log of bitcoin price on day t divided by bitcoin price on day $t-1$. *Crypto Ratio* is the number of comments that contain at least one technical word on day t divided by the total number of comments on day t . *Controls* contains *PCI Network*, ΔEPU , $\Delta Gold$, *SP500*, *SMB*, *HML*, *RMW*, and *CMA*. Control variables are defined in Appendix Table A1. Standard errors are bootstrapped using 1,000 replications. t statistics in parentheses * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

	Q10	Q30	Q50	Q70	Q90
<i>Crypto Ratio</i>	-1.315*** (-4.16)	-0.536** (-3.28)	-0.217 (-1.86)	-0.014 (-0.08)	0.121 (0.41)
<i>Constant</i>	-0.005 (-0.48)	0.004 (0.79)	0.007 (1.62)	0.014* (2.45)	0.038*** (3.73)
<i>F-stat</i>					4.57 0.001
<i>Controls</i>	Yes	Yes	Yes	Yes	Yes
<i>Quarterly FE</i>	Yes	Yes	Yes	Yes	Yes
<i>Pseudo R²</i>	0.086	0.050	0.039	0.040	0.050
<i>Obs</i>	1,510	1,510	1,510	1,510	1,510

Table 8: Equity Ratio OLS regressions

This table shows the association between daily SP500 returns and technical talk on equity subreddits. The dependent variable is the natural log of the SP500 index on day t divided by the SP500 index on day $t-1$. *Equity Ratio* is the number of comments that contain at least one technical word on day t divided by the total number of comments on day t . Other variables are defined in Appendix Table A1. Coefficients are estimated using Newey-West standard errors. t statistics in parentheses * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

	(1)	(2)	(3)
<i>Equity Ratio</i>	0.049 (1.54)	0.026 (0.85)	0.026 (0.83)
<i>SMB</i>		0.143 (1.93)	0.129 (1.72)
<i>HML</i>		0.224* (2.31)	0.246* (2.45)
<i>RMW</i>		-0.029 (-0.33)	-0.014 (-0.16)
<i>CMA</i>		-0.961*** (-6.42)	-0.995*** (-6.53)
<i>ΔEPU</i>			0.001 (1.46)
<i>ΔGold</i>			0.132* (2.19)
<i>Constant</i>	-0.001 (-0.72)	0.000 (0.24)	0.000 (0.24)
<i>Quarterly FE</i>	Yes	Yes	Yes
<i>Adj R²</i>	-0.001	0.103	0.112
<i>Obs</i>	1,510	1,510	1,510

Table 9: Equity ratio quantile regressions

This table shows the quantile regression results on the association between daily SP500 returns and technical talk on equity subreddits. The dependent variable is the natural log of the SP500 index on day t divided by the SP500 index on day $t-1$. *Equity Ratio* is the number of comments that contain at least one technical word on day t divided by the total number of comments on day t . *Controls* contains ΔEPU , $\Delta Gold$, $SP500$, SMB , HML , RMW , and CMA . Control variables are defined in Appendix Table A1. Standard errors are bootstrapped using 1,000 replications. t statistics in parentheses * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

	Q10	Q30	Q50	Q70	Q90
<i>Equity Ratio</i>	0.283*** (4.86)	0.097*** (4.35)	0.002 (0.09)	-0.095*** (-4.10)	-0.170*** (-3.45)
<i>Constant</i>	-0.019*** (-8.06)	-0.006*** (-5.90)	0.001 (1.19)	0.008*** (7.67)	0.019*** (10.20)
<i>F-Stat</i>					73.16 0.000
<i>Controls</i>	Yes	Yes	Yes	Yes	Yes
<i>Quarterly FE</i>	Yes	Yes	Yes	Yes	Yes
<i>Pseudo R²</i>	0.107	0.062	0.048	0.062	0.080
<i>Obs</i>	1,510	1,510	1,510	1,510	1,510

Table 10: Media outlet domains.

This table lists the domains for traditional media outlets (Panel A) and cryptocurrency media outlets (Panel B).

<i>Panel A: Traditional media domains</i>				
barrons.com	cnbc.com	independent.co.uk	nypost.com	thetstreet.com
bbc.co.uk	cnn.com	investing.com	nytimes.com	washingtonpost.com
bbc.com	finance.yahoo.com	marketwatch.com	reuters.com	wsj.com
bloomberg.com	forbes.com	money.cnn.com	sec.gov	
businessinsider.com	fortune.com	nasdaq.com	seekingalpha.com	
businesswire.com	ft.com	news.yahoo.com	theguardian.com	
<i>Panel B: Cryptocurrency media domains</i>				
ambcrypto.com	coindesk.com	cryptoslate.com	theblockcrypto.com	
beincrypto.com	cointelegraph.com	dailyhodl.com	themerke.com	
bitcoinist.com	cryptobriefing.com	decrypt.co	u.today	
bitcoinmagazine.com	cryptocoinsnews.com	news.bitcoin.com		
blockworks.co	cryptonews.com	newsbtc.com		
coinbureau.com	cryptopotato.com	theblock.co		

Table 11: Crypto Ratio and News (Quantile)

This table shows the quantile regression results on the association between daily news and technical talk on cryptocurrency subreddits. The dependent variables are proxies for attention to news from traditional and cryptocurrency related media outlets. *Traditional News* (*Crypto News*) is the number of comments on submissions referencing a media article with a domain listed in Table 10 Panel A (Panel B) on day t divided by the total number of comments on day t . *Abn Traditional News* (*Abn Crypto News*) is the natural log of the number of comments on submissions referencing a media article with a domain listed in Table 10 Panel A (Panel B) on day t minus its 30-day moving average. *Crypto Ratio* is the number of comments that contain at least one technical word on day t divided by the total number of comments on day t . The models include control variables *Bitcoin*, *PCI Network*, ΔEPU , $\Delta Gold$, *SP500*, *SMB*, *HML*, *RMW*, and *CMA*. Control variables are defined in Appendix Table A1. Standard errors are bootstrapped using 1,000 replications. t statistics in parentheses * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

	Crypto News	Abn Crypto News	Traditional News	Abn Traditional News	Total News	Abn Total News
<i>Q10</i>						
<i>Crypto Ratio</i>	0.024 (0.48)	4.684 (1.28)	0.120*** (4.62)	18.561** (2.70)	0.119 (1.20)	3.680 (1.09)
<i>Q30</i>						
<i>Crypto Ratio</i>	0.006 (0.07)	-0.256 (-0.09)	0.203*** (3.65)	3.293 (0.71)	0.056 (0.33)	0.138 (0.05)
<i>Q50</i>						
<i>Crypto Ratio</i>	-0.089 (-0.67)	-7.434** (-2.94)	0.229* (2.44)	-2.711 (-0.63)	-0.034 (-0.18)	-3.381 (-1.27)
<i>Q70</i>						
<i>Crypto Ratio</i>	-0.110 (-0.49)	-9.783** (-3.21)	0.093 (0.77)	-9.460* (-1.99)	-0.039 (-0.12)	-4.705 (-1.55)
<i>Q90</i>						
<i>Crypto Ratio</i>	-0.135 (-0.49)	-13.166** (-3.22)	-0.058 (-0.21)	-15.766* (-2.49)	0.291 (0.78)	-11.936** (-2.84)
<i>Obs</i>	1,510	1,510	1,510	1,510	1,510	1,510

Appendix

Table A1: Variable definitions

Variable	Description
<i>Crypto Ratio</i>	The number of comments in r/CryptoCurrency and r/Bitcoin that contain at least one technical word on day t divided by the total number of comments in r/CryptoCurrency and r/Bitcoin on day t.
<i>Equity Ratio</i>	The number of comments in r/wallstreetbets, r/stocks, and r/investing that contain at least one technical word on day t divided by the total number of comments in r/wallstreetbets, r/stocks, and r/investing on day t.
<i>Bitcoin</i>	Daily bitcoin return is calculated as the natural log of the price of bitcoin on day t divided by the price of bitcoin on day t-1.
<i>SP500</i>	Daily S&P 500 return is calculated as the natural log of the S&P 500 index on day t divided by the S&P 500 index on day t-1.
<i>ΔTransactions</i>	Number of confirmed transactions on Bitcoin's blockchain on day t from https://www.blockchain.com/explorer/charts . We calculate the daily change in transactions as the natural log of the value on day t divided by the value on day t-1.
<i>ΔUnique Addresses</i>	Total number of unique addresses used on Bitcoin's blockchain on day t from https://www.blockchain.com/explorer/charts . We calculate the daily change in unique addresses as the natural log of the value on day t divided by the value on day t-1.
<i>ΔTransaction Value</i>	Estimated transaction value in bitcoin on the blockchain on day t from https://www.blockchain.com/explorer/charts . We calculate the daily change in transaction value as the natural log of the value on day t divided by the value on day t-1.
<i>ΔUnspent Transactions</i>	Number of unspent Bitcoin transaction outputs from https://www.blockchain.com/explorer/charts . We calculate the daily change in unspent Bitcoin transactions as the natural log of the value on day t divided by the value on day t-1.
<i>ΔTransactions perSecond</i>	Average number of Bitcoin transactions added to the mempool per second on day t from https://www.blockchain.com/explorer/charts . We calculate the daily change in transactions per second as the natural log of the value on day t divided by the value on day t-1.
<i>ΔMempool Size</i>	Total size of transactions (in bytes) waiting to be confirmed on day t from https://www.blockchain.com/explorer/charts . We calculate the daily change in mempool size as the natural log of the value on day t divided by the value on day t-1.
<i>ΔMempool Count</i>	Number of valid transactions waiting to be confirmed on day t from https://www.blockchain.com/explorer/charts . We calculate the daily change in mempool count as the natural log of the value on day t divided by the value on day t-1.
<i>PCI Network</i>	The first component from PCA on number of transactions, number of unique addresses, estimated transaction value, unspent bitcoin transactions, number of transactions per second, mempool size, and mempool count.
<i>ΔEPU</i>	News-based Economic Policy Index (EPU) from https://www.policyuncertainty.com/us_monthly.html . We calculate the daily change in EPU as the natural log of EPU on day t divided by EPU on day t-1.
<i>ΔVIX</i>	Daily CBOE Volatility Index from Federal Reserve Bank of St. Louis (FRED) https://fred.stlouisfed.org/series/VIXCLS . We calculate the daily change in VIX as the natural log of VIX on day t divided by VIX on day t-1.
<i>ΔGold</i>	Daily S&P GSCI Gold Spot from Refinitiv DataStream. We calculate the daily change in Gold as the natural log of Gold on day t divided by Gold on day t-1.

Table A2: Sample of predicted words

Panel A: Sample of predicted words used				
deviation band	vwap standard	squeeze indicator	momentum indicator	volume weight
awesome oscillator	use dmi	indicator stochastic	macd ema	indicator ma
slow stochastic	auto trend	auto fib	rsi standard	midline
upward slope	bband	dev band	strength indicator	average crossover
sd channel	upward stochastic	average rsi	williams alligator	anchor vwap
fibonacci extension	ewo	rsi stochastic	bar replay	rsi mfi
stdev	automatically plot	directional simple	line vwap	fibo
market cap	macd etc	rsi bolling	smi	vwap std
standard dev	macd histogram	trend channel	vwma	smas emas
ma macd	mfi rsi	channel draw	fibonacci level	stochastic etc
Panel B: Sample of predicted words removed				
cross red	smas	typically indicate	identify key	main indicator
red blue	hand draw	generally reliable	zone create	lambda
cross over	diagonal	support amp	major swing	change length
analyzer	drawing tool	previous period	true ta	price
fix range	connect peak	differently different	determine support	average
dot line	zigzag	easy identify	mark important	supply dry
ichi	brooks youtube	pattern etc	tema	buyer willing
calculate differently	window screen	map key	stoch	purchase price
modified	sha	use horizontal	ao	current price
require custom	vp chart	look divergence	amp ma	show majority